PUBLICATION-READY REPORT (Paper Version)
Geotechnical ML Prediction Tool

Model Mapping Summary:
PHI: Paper=Extra Trees | Tool=Extra Trees
SP_GR: Paper=CatBoost | Tool=CatBoost
COHESION: Paper=Extra Trees | Tool=Extra Trees
MC: Paper=CatBoost | Tool=CatBoost

BENCHMARK RATIONALE

- Eleven representative machine learning algorithms were selected to span four major regression families: linear (Linear Regression, Ridge, Lasso), kernel-based (Support Vector Regression), a single decision tree, bagging ensembles (Random Forest, Extra Trees), and boosting ensembles (Gradient Boosting, XGBoost, LightGBM, CatBoost). These families were included to represent a broad range of model complexities and inductive biases relevant to nonlinear, tabular geotechnical data. For each target property, all eleven algorithms were benchmarked using the same borehole-group-aware train/test split, and the best-ranked algorithm for each target property was selected using test-set RMSE and MAE (rank-averaged), with |MBE| used only as a bias-check tie-breaker and Group-CV R2 kept as supporting robustness evidence -- that best-ranked algorithm was then used for deployment in the final prediction tool. The full eleven-model benchmark is provided as supplementary material.

ENGINEERING INTERPRETATION

- MOISTURE CONTENT(%) had the lowest mean Test R2 across the eleven models (0.498), indicating it was the most difficult property to predict from the available predictors.
- COHESION(KN/M2) had the highest mean Test R2 across the eleven models (0.761), indicating it was the most reliably predicted property.
- For Angle of Friction,ϕ, Extra Trees gave the best fit under the RMSE/MAE rank-average selection criterion (RMSE = 1.843, MAE = 1.351, MBE = -0.202 [bias check]; Test R2 = 0.835 and Group-CV R2 = 0.813 reported as supporting evidence), which may be associated with its randomized tree-ensemble structure, which can represent nonlinear relationships and interactions among the predictors while reducing variance.
- For Specific Gravity, CatBoost gave the best fit under the RMSE/MAE rank-average selection criterion (RMSE = 0.085, MAE = 0.060, MBE = -0.014 [bias check]; Test R2 = 0.617 and Group-CV R2 = 0.416 reported as supporting evidence), which may be associated with its ordered-boosting strategy, which can capture nonlinear interactions among the numerical SPT/depth-based predictors and the one-hot encoded soil-type indicators.
- For Cohesion(KN/m2), Extra Trees gave the best fit under the RMSE/MAE rank-average selection criterion (RMSE = 4.137, MAE = 1.459, MBE = -0.323 [bias check]; Test R2 = 0.839 and Group-CV R2 = 0.828 reported as supporting evidence), which may be associated with its randomized tree-ensemble structure, which can represent nonlinear relationships and interactions among the predictors while reducing variance.
- For Moisture Content(%), CatBoost gave the best fit under the RMSE/MAE rank-average selection criterion (RMSE = 2.992, MAE = 1.929, MBE = +0.468 [bias check]; Test R2 = 0.589 and Group-CV R2 = 0.475 reported as supporting evidence), which may be associated with its ordered-boosting strategy, which can capture nonlinear interactions among the numerical SPT/depth-based predictors and the one-hot encoded soil-type indicators.

LIMITATION

- All performance metrics are based on a group-aware train-test split per target; results should be treated as indicative.
- The residual bootstrap uncertainty band reported for each target is an in-sample diagnostic -- it is constructed from, and its coverage checked against, the same holdout residuals -- and should not be interpreted as an independently validated 95% prediction interval.
- The training dataset contained no laboratory Angle of Friction (phi) or Cohesion values for Clay(CL) samples; the deployed prediction tool therefore does not predict phi or cohesion for Clay(CL), while Specific Gravity and Moisture Content remain available for that soil type.